

EA Parameter Quick Reference Guide

Parameter Categories

Symbol & Identity

Parameter	Type	Default	Range	Description
MagicNumber	int	100140	1-999999	Unique identifier for this EA's orders. Change if running multiple EAs.

Lots & Grid

Parameter	Type	Default	Range	Description
LotSize	double	0.01	0.01-100	Initial lot size for first trade and grid levels
UseFixedLot	bool	true	true/false	If true, all grid levels use same lot. If false, allows progression.
MaxGridLevels	int	10	1-50	Maximum number of grid levels (orders) in basket
GridDistancePoints	int	500	100-5000	Distance in points between grid levels
UseDynamicGrid	bool	true	true/false	Enable dynamic grid expansion during stress
DynamicGridMultiplier	double	1.5	1.0-3.0	Multiplier for grid distance when DD threshold reached
MaxLotMultiplier	double	1.0	1.0-3.0	Maximum lot multiplier (1.0 = no progression)

Notes:

- 1 point = 0.01 for XAUUSD on most brokers (check your broker's point size)
- 500 points = \$5 move in gold price
- Dynamic grid helps survive strong trends by spacing out new levels

Entry (Phase-1: Manual and Semi-Auto Only)

Parameter	Type	Default	Range	Description
EntryMode	int	0	0-1	Phase-1: 0 = Manual (adopt user order), 1 = Semi-Auto (user confirms via input)
OpenBasketDirection	int	0	0-2	Semi-Auto only: 0 = None, 1 = BUY, 2 = SELL (confirm to open basket)

Parameter	Type	Default	Range	Description
<code>SuggestedDirectionSource</code>	int	0	0-1	Semi-Auto display: 0 = None, 1 = MA (suggestion only; no auto-open)
<code>MA_Period</code>	int	50	5-200	MA period (for suggestion display only in Phase-1)
<code>RSI_Period</code>	int	14	5-50	RSI period (Phase-2; not used for entry in Phase-1)
<code>RSI_Oversold</code>	double	30	10-40	RSI oversold (Phase-2)
<code>RSI_Overbought</code>	double	70	60-90	RSI overbought (Phase-2)
<code>SessionStartHour</code>	int	0	0-23	Session start hour (Phase-2)

Phase-1 Entry Mode Details:

- **Mode 0 (Manual – Adopt):** EA does **not** open the first trade. You place order(s) in MT4 with the EA's MagicNumber; the EA adopts them as the basket start and manages grid/recovery. Single direction only.
- **Mode 1 (Semi-Auto):** EA can show a **suggested** direction (e.g. from MA if `SuggestedDirectionSource` = 1). You set `OpenBasketDirection` to BUY or SELL to confirm; when filters pass, the EA opens the first trade in that direction.

Phase-2 (future): Indicator-based entry (MA/RSI/Time) is out of scope for Phase-1; code is present for later phases (EntryMode 2=MA, 3=RSI, 4=Time).

🔗 Exit

Parameter	Type	Default	Range	Description
<code>BasketTakeProfitMoney</code>	double	10.0	1.0-10000	Basket TP in account currency (e.g., USD)
<code>UseTrailingBasketTP</code>	bool	false	true/false	Enable trailing basket take profit
<code>TrailingStart</code>	double	15.0	5.0-10000	Profit level where trailing activates
<code>TrailingStep</code>	double	5.0	1.0-1000	Trail by this amount (closes if profit drops by this)

Example:

- `BasketTakeProfitMoney` = 10.0 → closes basket when profit reaches \$10
- With trailing: if profit reaches \$15, then drops to \$10 (\$15 - \$5 step), basket closes

⚠ Risk

Parameter	Type	Default	Range	Description
MaxBasketDrawdownMoney	double	100.0	10-10000	Maximum basket DD in account currency (reference for % calculations)
MaxEquityDrawdownPercent	double	20.0	5-50	Maximum account equity DD % (blocks new baskets)
PauseNewTradesAtDDPercent	double	50.0	10-90	Basket DD % to pause grid expansion
IncreaseGridDistanceAtDDPercent	double	30.0	10-80	Basket DD % to activate dynamic grid expansion
EmergencyCloseAtDDPercent	double	80.0	50-100	Basket DD % to trigger emergency close
MinFreeMarginPercent	double	30.0	10-80	Minimum free margin % required

Risk Flow (ProtectionMode / EmergencyExit):

- 1. At 30% basket DD → **ProtectionMode**: dynamic grid activates (if enabled)
- 2. At 50% basket DD → **ProtectionMode**: grid expansion pauses
- 3. At 80% basket DD → **EmergencyExit**: controlled close triggered (no blind liquidation)

Important:

- MaxBasketDrawdownMoney is the reference for calculating basket DD%
- If basket loss = \$50 and MaxBasketDrawdownMoney = 100, basket DD = 50%

 Emergency (EmergencyExit)

Parameter	Type	Default	Range	Description
UseEmergencyExit	bool	true	true/false	Enable EmergencyExit (controlled, configurable; minimizes damage)
MaxBasketSlippagePoints	int	50	10-200	Maximum slippage allowed when closing basket

EmergencyExit Triggers:

- Basket DD ≥ EmergencyCloseAtDDPercent
- Equity DD ≥ MaxEquityDrawdownPercent
- Free margin < 50% of MinFreeMarginPercent

 Filters

Parameter	Type	Default	Range	Description
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Parameter	Type	Default	Range	Description
MaxSpreadPoints	int	50	10-200	Maximum spread allowed for trading (points)
UseVolatilityFilter	bool	true	true/false	Enable ATR-based volatility filter
ATR_Period	int	14	5-50	ATR calculation period
ATR_MultiplierThreshold	double	2.0	1.5-5.0	ATR multiplier threshold for high volatility
BlockNewBasketAboveVolatility	bool	true	true/false	Block new basket creation during high volatility
PauseGridAboveVolatility	bool	true	true/false	Pause grid expansion during high volatility

Volatility Logic:

- EA calculates 20-period average ATR
- If current ATR > (average ATR × ATR_MultiplierThreshold), volatility is "high"
- High volatility blocks new baskets and/or pauses grid (based on settings)

⌚ Session

Parameter	Type	Default	Range	Description
UseTradingHours	bool	false	true/false	Enable trading hours filter
StartHour	int	0	0-23	Trading start hour (broker server time)
EndHour	int	23	0-23	Trading end hour (broker server time)

Notes:

- Hours are in broker server time (check MT4 Market Watch time)
- If StartHour > EndHour, assumes overnight session (e.g., 22-6)
- Filter only blocks NEW basket creation, not management of existing basket

Preset Configurations

🌀 Conservative (Low Risk)

```
LotSize = 0.01
MaxGridLevels = 5
GridDistancePoints = 800
BasketTakeProfitMoney = 5.0
MaxBasketDrawdownMoney = 50.0
MaxEquityDrawdownPercent = 10.0
PauseNewTradesAtDDPercent = 40.0
```

```
IncreaseGridDistanceAtDDPercent = 25.0  
EmergencyCloseAtDDPercent = 70.0  
MinFreeMarginPercent = 40.0  
MaxSpreadPoints = 30
```

Best for: Small accounts, beginners, high-risk aversion

⦿ Moderate (Balanced)

```
LotSize = 0.01  
MaxGridLevels = 10  
GridDistancePoints = 500  
BasketTakeProfitMoney = 10.0  
MaxBasketDrawdownMoney = 100.0  
MaxEquityDrawdownPercent = 20.0  
PauseNewTradesAtDDPercent = 50.0  
IncreaseGridDistanceAtDDPercent = 30.0  
EmergencyCloseAtDDPercent = 80.0  
MinFreeMarginPercent = 30.0  
MaxSpreadPoints = 50
```

Best for: Standard accounts, tested strategies, moderate risk tolerance

⦿ Aggressive (Higher Risk)

```
LotSize = 0.02  
MaxGridLevels = 15  
GridDistancePoints = 300  
BasketTakeProfitMoney = 20.0  
MaxBasketDrawdownMoney = 200.0  
MaxEquityDrawdownPercent = 30.0  
PauseNewTradesAtDDPercent = 60.0  
IncreaseGridDistanceAtDDPercent = 40.0  
EmergencyCloseAtDDPercent = 90.0  
MinFreeMarginPercent = 25.0  
MaxSpreadPoints = 80
```

Best for: Larger accounts, experienced traders, higher risk tolerance

⚠ **Warning:** Aggressive settings can lead to significant drawdowns. Only use if you understand the risks.

Optimization Tips

For Ranging Markets

- Decrease `GridDistancePoints` (300-400)
- Increase `MaxGridLevels` (12-15)
- Lower `BasketTakeProfitMoney` for quicker exits

For Trending Markets

- Increase `GridDistancePoints` (600-1000)
- Decrease `MaxGridLevels` (5-8)
- Enable `UseDynamicGrid` with higher multiplier (2.0-2.5)
- Tighten `PauseNewTradesAtDDPercent` (30-40%)

For Volatile Markets (News Events)

- Increase `MaxSpreadPoints` (80-100) or avoid trading
- Enable `UseVolatilityFilter = true`
- Lower `ATR_MultiplierThreshold` (1.5-1.8) to be more sensitive
- Use `UseTradingHours` to avoid news times

For Low Margin Accounts

- Decrease `LotSize` (0.01 or less)
- Decrease `MaxGridLevels` (5-7)
- Increase `MinFreeMarginPercent` (40-50%)
- Tighten `MaxEquityDrawdownPercent` (10-15%)

Common Parameter Mistakes

✗ Too many grid levels with small distance

- Problem: Rapid grid filling, high exposure
- Fix: Either increase `GridDistancePoints` OR decrease `MaxGridLevels`

✗ Emergency close too close to pause level

- Problem: Emergency triggers before pause has effect
- Fix: Ensure `EmergencyCloseAtDDPercent` is at least 20-30% higher than `PauseNewTradesAtDDPercent`

✗ Basket TP smaller than expected DD

- Problem: Basket rarely reaches TP, often hits DD limits
- Fix: Increase `BasketTakeProfitMoney` or tighten grid to reduce DD

✗ Spread filter too tight

- Problem: EA never trades due to spread rejection
- Fix: Check typical XAUUSD spread on your broker, set `MaxSpreadPoints` accordingly (usually 30-80)

✗ Lot size too large for account

- Problem: Margin call risk, emergency closes

- Fix: Use 0.01 lot per \$1000-2000 account balance as starting point
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Testing Checklist

Before going live, verify these scenarios in Strategy Tester:

- ☐ Basket opens correctly in both directions (BUY and SELL)
 - ☐ Grid levels add at correct distances
 - ☐ Dynamic grid activates at DD threshold
 - ☐ Grid pauses at pause threshold
 - ☐ Basket closes at TP
 - ☐ Emergency close triggers at DD limit
 - ☐ Spread filter blocks trades during high spread
 - ☐ Volatility filter blocks trades during high ATR
 - ☐ Trading hours filter works correctly
 - ☐ Margin check prevents over-leveraging
 - ☐ EA recovers state after restart with open positions
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Remember: Start conservative, test thoroughly, and only increase risk after confirming stable behavior.